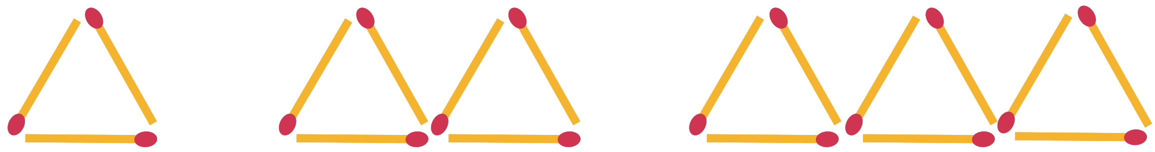
Geometric patterns

1 Yvonne has constructed the first three triangles of a pattern using matchsticks:



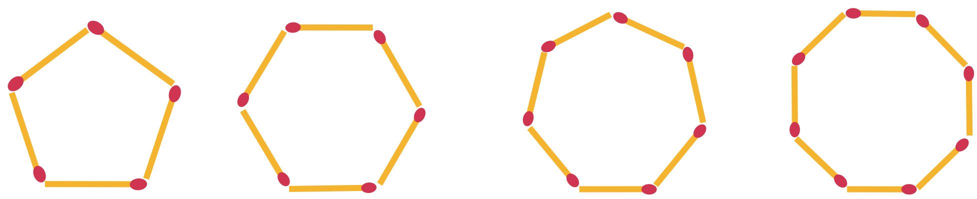
a Find the number of matchsticks that are used to make each new triangle.

b Describe the relationship between the number of triangles and the number of matchsticks Yvonne will need for his pattern.

c Write the algebraic rule for the number of matchsticks, M, in terms of the number of triangles, T, for this pattern.

2 Peter is making a sequence of shapes out of matchsticks:

Figure 1 Figure 2 Figure 3 Figure 4



Peter makes a table comparing the figure number to the number of matchsticks needed to construct it as shown:

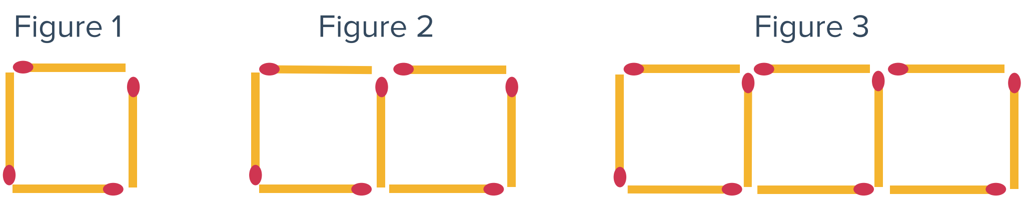
Figure number	1	2	3	4
Matchsticks	5	6		

a Complete the table for the above pattern.

b Describe the relationship between the figure number and the number of matchsticks used to make it.

c Write the algebraic rule for the number of matchsticks, M, in terms of the Figure number, F, for this pattern.

3 Dave is constructing a continuing pattern [redacted]ares using matchsticks:



a If Dave wanted to continue the pattern, determine the number of matchsticks he would need for each square he adds.

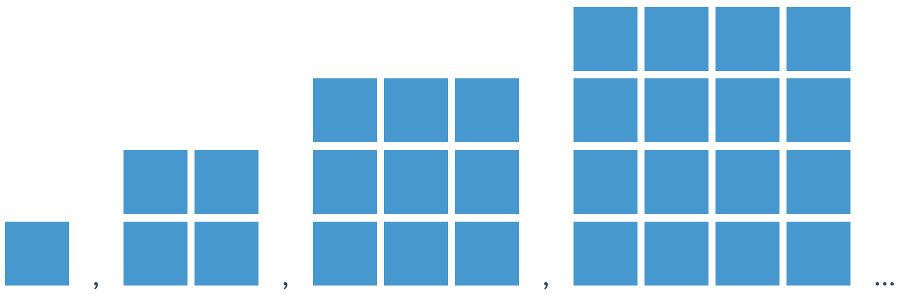
b Dave made a table comparing the figure number to the number of matchsticks required. Describe any patterns that you notice.

Figure no.	1	2	3	4
Matchsticks	4	7	10	13

c Describe the relationship between the figure number and the number of matchsticks it requires.

d Write the algebraic rule for the number of matchsticks, M, in terms of the Figure number, S, for this pattern.

4 Vanessa makes the first four entries of a sequence out of coloured tiles and as shown:



She then creates a table to show the relationship between the entry number and the number of tiles used for that entry:

Entry number (E)	1	2	3	4
No. of tiles used (T)	1			

a Complete the table of values.

b Describe a rule that Vanessa could use to find the number of tiles required for each entry.

c Write the algebraic rule for the number of tiles, T, in terms of the entry number, E, for this pattern.

Equations and tables

5 Write an equation for y in terms of x for the following rules:

a "The value of y is six less than the value of x".

b "The value of y is three times the value of x".

c "The value of y is five more than two times x".

6 For each of the following tables:

i Describe the relationship between x and y.

ii Write an equation for y in terms of x.

a

x	1	2	3	4	5
y	9	10	11	12	13

b

x	3	6	9	12	15
y	1	2	3	4	5

c

x	6	7	8	9	10
y	1	2	3	4	5

d

x	1	2	3	4	5
y	7	14	21	28	35

e

x	1	2	3	4	5
y	7	12	17	22	27

f

x	1	2	3	4	5
y	14	11	8	5	2

7 For each table of values below, write an equation for y in terms of x:

a

x	6	7	8	9	10
y	3	4	5	6	7

b

x	10	11	12	13	14
y	50	55	60	65	70

c

x	6	7	8	9	10
y	13	14	15	16	17

d

x	5	6	7	8	9
y	12	14	16	18	20

e

x	5	6	7	8	9
y	2	4	6	8	10

f

x	72	81	90	99	108
y	8	9	10	11	12

8 For each equation below, complete a tabl[redacted]alues of the form:

x	1	2	3	4
y				

a $y = 7x$

b $y = x + 8$

c $y = 2x - 6$

d $y = \dfrac{x}{4}$

e $y = 5x + 1$

f $y = -3x$

g $y = \dfrac{2x}{3}$

h $y = 10 - 4x$

Equations for relationships

9 Gwen sells bananas in bunches of 4.

a Complete the following table:

Bunches, x	1	2	3	4
Bananas, y				

b Describe the relationship between the number of bunches and the number of bananas.

c Write an equation for the number of bananas, y, in terms of the number of bunches, x.

10 Quentin buys some decks of playing cards that contain 52 cards each.

a Complete the following table:

Decks, d	1	2	3	4
Cards, c				

b Describe the relationship between the number of decks and the number of cards.

c Write an equation for the number of cards, c, in terms of the number of decks, d.

11 Vanessa opens a bank account and depos[illegible]00. At the end of each week she adds $\$10$ to her account.

a Complete the following table which shows the balance of Vanessa's account over the first four weeks:

Week (W)	0	1	2	3	4
Account total (A)	$\$300$	$\$310$			

b Write the algebraic rule for Vanessa's account total, A, in terms of the number of weeks W, for which she has been adding to her account.

12 Quentin already owns 5 marbles. He then buys some bags of marbles containing 4 marbles each.

a Describe the relationship between the number of marbles Quentin will have in total and the number of bags of marbles he buys.

b Write the algebraic rule for the number of marbles Quentin will own, y, in terms of the number of bags he buys, x.